Housing Discrimination and Algorithmic Bias: A Data Ethics Project

Background and Context

Historical Housing Discrimination

Racial covenants were legal clauses in property deeds that prohibited sale or occupancy by non-white people. These discriminatory practices, along with redlining and other systematic forms of housing discrimination, shaped the demographic and economic patterns we see in our cities today. While racial covenants were made illegal by the Fair Housing Act of 1968, their effects continue to influence modern housing patterns, property values, and wealth distribution.

The Mapping Prejudice Project at the University of Minnesota has documented over 30,000 racial covenants in Hennepin County. These covenants, implemented between 1910 and 1955, created lasting patterns of segregation and economic disparity that persist in modern data.

Modern Mortgage Lending

Today, most mortgage lending decisions involve algorithmic systems that assess credit risk and determine loan approval. While these systems don't explicitly consider race, they may perpetuate historical biases through: - Property valuations influenced by historical segregation - Neighborhood characteristics shaped by past discrimination - Economic factors that reflect generational wealth disparities

Project Overview

This project examines the relationship between historical housing discrimination and modern lending patterns, focusing on two distinct types of bias:

1. **Data Bias**: How historical discrimination is embedded in modern data
   - Property values in areas with/without historical covenants
   - Neighborhood demographic and economic characteristics
   - Patterns of generational wealth and investment
2. **Algorithmic Bias**: How modern lending systems might perpetuate discrimination
   - Mortgage approval rates and terms
   - Risk assessment criteria
   - The use of potentially biased proxy variables

Data Sources

Primary Dataset (Provided)

You will receive a preprocessed dataset combining: - Covenant density by census tract - Property values - Modern mortgage lending data - Census demographic information

Original Data Sources (For Reference and Extension Work)

1. **Mapping Prejudice Project**
   - URL: https://mappingprejudice.umn.edu/
   - Contains: Historical covenant data for Hennepin County
   - Format: GIS data showing covenant locations and details
2. **Census TIGER/LINE Files**
   - URL: https://www.census.gov/cgi-bin/geo/shapefiles/index.php?year=2024&layergroup=Census+Tracts
   - Contains: Census tract boundaries
   - Format: Geographic shapefiles
3. **HMDA Data Browser**
   - URL: https://ffiec.cfpb.gov/data-browser/data/2023?category=states&items=MN
   - Contains: Modern mortgage lending data
   - Format: CSV files with lending information

Project Components

1. Initial Analysis

Using the provided dataset: - Analyze relationships between covenant density and property values - Examine modern mortgage approval patterns - Identify potential proxy variables for historical discrimination - Compare lending patterns in historically covenanted vs. non-covenanted areas

2. Bias Investigation

Clearly distinguish between and analyze: - **Data Bias**: Document how historical discrimination appears in modern data - Property value disparities - Neighborhood characteristics - Economic indicators - **Algorithmic Bias**: Examine how lending algorithms might perpetuate discrimination - Approval rate patterns - Loan term differences - Risk assessment criteria

3. Ramsey County Extension

Extend the analysis to Ramsey County: 1. Gather data from the provided sources 2. Adapt the analysis methods from Hennepin County 3. Compare patterns between counties 4. Document challenges and limitations

Deliverables

GitHub Repository

- Analysis code and documentation
- Data processing scripts
- Meeting summaries (AI-assisted)
- Visualization code
- README explaining repository organization

Final Report (5 pages)

1. Introduction and Historical Context
2. Methodology
3. Findings
    - Data bias analysis
    - Algorithmic bias analysis
    - County comparison
4. Discussion
    - Implications for modern lending
    - Proposed mitigation strategies
5. Limitations and Future Work

Presentation (15 minutes)

- Historical context
- Methodology overview
- Key findings
- Visualizations
- Recommendations

Technical Resources

- You will receive access to:
    - Preprocessed dataset for Hennepin County
    - Python notebook showing data preparation steps
    - Sample visualization code

Evaluation

Grading will follow the provided rubrics, with special emphasis on: - Clear distinction between data and algorithmic bias - Quality of technical analysis - Depth of historical understanding - Effectiveness of communication - Success in Ramsey County extension

Tips for Success

- Start with the provided dataset to understand basic patterns
- Document all assumptions and methodological choices
- Clearly separate findings related to data bias vs. algorithmic bias
- Consider ethical implications throughout your analysis
- Plan the Ramsey County extension early - data collection takes time

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[Previous sections remain the same through "Technical Resources"]

Project Steps

Step 1: Set Up Project Infrastructure

1. **Create the GitHub Repository**
   - One group member creates a repository named **DATA6550-HOUSING-BIAS**
   - Required directory structure:

```
FINAL_REPORT.PDF
README.MD
CODE/
  LASTNAME1/
  LASTNAME2/
  LASTNAME3/
  LASTNAME4/
DATA/
  HENNEPIN_ANALYSIS.CSV
  [ADDITIONAL DATASETS]
COLLABORATION/
  WEEKA.PDF
  WEEKB.PDF
ANALYSIS/
  [GRAPHICS AND INTERMEDIATE RESULTS]
```

   - Share repository access with all group members
   - Add the instructor as a collaborator

2. **Set Up Shared Document**
   - Create a shared Google document for the final report of this project. Google Docs will automatically record all document updates and it is one of the ways the instructor will track your contributions. Share the document with the team, the instructor, and add a link to it in the Dropbox for the project in D2L along with uploading a PDF of the document. My Gmail account is JFWALLIN@GMAIL.COM when you add me to the document (with editing rights, please). Please submit only one PDF and link per group.

Step 2: Create Group Communication Plan

1. Choose a communication platform (D2L forums, Teams, Zoom, etc.)
2. Invite all group members and instructor to the platform
3. Create AI summaries of all meetings
4. Establish regular check-in schedule
5. Document communication protocols in repository README

Step 3: Initial Data Exploration

1. Download and examine the provided Hennepin County dataset
2. Review documentation of data sources:
    - Mapping Prejudice Project
    - Census TIGER/LINE files
    - HMDA Data Browser
3. Document initial observations in shared report

Step 4: Divide and Execute Tasks

1. Assign specific analysis responsibilities:
    - Historical context research
    - Data bias analysis
    - Algorithmic bias investigation
    - Visualization development
2. Each member must contribute their own code to their personal directory
3. Regular code reviews and integration discussions

Step 5: Document Discussions

1. After each group meeting, create AI-generated summary including:
    - Key decisions and insights
    - Task assignments and deadlines
    - Challenges and solutions
    - You can add a supplemental document or text if the AI summary does not capture your discussion.
2. Save summaries as WeekA.pdf and WeekB.pdf in Collaboration folder. You may add additional documents if the AI summary doesn't capture the collaboration.

Step 6: Conduct Primary Analysis

1. Analyze Hennepin County data:
    - Property value patterns
    - Lending disparities
    - Demographic correlations
2. Document methodology and findings
3. Create visualizations
4. Regular repository updates

Step 7: Ramsey County Extension

1. Collect required datasets
2. Apply analysis methodology
3. Compare results with Hennepin County
4. Document challenges and limitations

Step 8: Complete Final Report

1. Compile findings in shared Google document
2. Include sections on:

  - Historical context
  - Methodology
  - Results
  - Comparative analysis
  - Recommendations
3. Ensure all members contribute with tracked changes

Step 9: Submit Deliverables

1. **GitHub Repository**
  - Complete code base
  - Documentation
  - Analysis results
  - Meeting summaries
2. **D2L Group Dropbox**
  - Final report PDF
  - Collaboration summaries - WeekA.pdf, WeekB.pdf, and supplemental document if needed
  - Repository URL
  - Presentation slides
3. **Individual Dropbox**
  - Personal contribution summary
  - Lessons learned
  - Reflection on findings

Step 10: Present Findings

1. Prepare a 5-minute presentation
2. Include:
  - Historical context
  - Methodology overview
  - Key findings
  - Visualizations
  - Recommendations
3. Be prepared for Q&A

[Previous sections on Evaluation remain the same]

Important Notes

- Submit only one repository URL and report per group
- Maintain regular commits to show consistent progress
- Document all data sources and methodological decisions
- Clearly distinguish between data bias and algorithmic bias findings
- Start Ramsey County data collection early
- Use course forums or office hours for technical support